

Supplementary Material for

**“How Much Fluctuation Is Dangerous?
Predicting Tipping Points via Statistically Conditioned
Digital Twins”**

Detailed model equations, statistical coordinates, training protocols,
reservoir settings, outcome criteria, and numerical results

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S1 Scope and notation

This Supplementary Material records the numerical protocol used for the final food-chain, voltage-collapse, and Kuramoto experiments. It complements the main text by specifying the target equations, conditioning statistics, data windows, reservoir implementation, test coordinates, outcome rules, and numerical thresholds. The physical bifurcation parameter is used to generate and validate ground-truth trajectories, but it is not supplied to the SCDT during training or autonomous prediction.

For a sampled trajectory, n denotes the discrete sample index and Δt the sampling interval. Three different initialization intervals must be kept distinct:

- **Target transient:** the target ODE is integrated before the retained record begins. These samples are discarded and are never supplied to the reservoir.
- **Training washout:** retained target samples are supplied to the reservoir, but the early reservoir states are excluded from the readout regression so that dependence on the zero reservoir state is reduced.
- **Prediction warm-up (teacher forcing):** a target segment is supplied immediately before autonomous prediction. These samples initialize the reservoir state; the subsequent SCDT output is fed back as its next input.

The labels *inter*, *pre*, and *post* denote, respectively, a test point inside the training range, a test point beyond the training range but before the physical transition, and a point beyond the transition.

S2 Statistically conditioned reservoir

S2.1 Reservoir update

Let $\mathbf{u}_n \in \mathbb{R}^D$ be the dynamical input, $\mathbf{r}_n \in \mathbb{R}^{N_r}$ the reservoir state, and $\mathbf{s} \in \mathbb{R}^2$ a pair of statistics that remains constant over one trajectory segment. After the system-specific normalization $\mathcal{T}_s$, the reservoir evolves as

$$\mathbf{r}_{n+1} = (1 - \alpha)\mathbf{r}_n + \alpha \tanh(A\mathbf{r}_n + W_{\text{in}}\mathbf{u}_n + W_{\text{stat}}\mathcal{T}_s(\mathbf{s})), \quad (1)$$

where A and the input matrices are fixed after random initialization, α is the leakage rate, and only the output matrix is trained. The statistics channel therefore acts as a trajectory-level bias that indexes a family of reservoir dynamics.

For one-step training, the output weights solve

$$W_{\text{out}} = \arg \min_W \sum_{m=1}^M \sum_{n \in \mathcal{I}_m} \left\| W \mathbf{z}_{n+1}^{(m)} - \mathbf{u}_{n+1}^{(m)} \right\|_2^2 + \beta \|W\|_F^2, \quad (2)$$

where $M = 5$ is the number of training conditions, $\mathcal{I}_m$ excludes the washout interval, β is the ridge coefficient, and $\mathbf{z}$ is the system-specific readout feature in Table 1. During autonomous prediction, $\hat{\mathbf{u}}_{n+1} = W_{\text{out}}\mathbf{z}_{n+1}$ is fed back as $\mathbf{u}_{n+1}$.

Table 1. System-specific dynamical variables, statistic normalization, and readout feature. These details refine the generic readout equation in the main text.

System	Dynamic input/output	Statistic transform $\mathcal{T}_s$	Readout feature $\mathbf{z}$
Food chain	$[R, C, P]^\top$	Training-set z-score, followed by elementwise gain (0.1, 0.1) and bias (-1.63179994, 0.05175686)	$\mathbf{z} = \phi(\mathbf{r})$, where zero-based odd entries are squared: $\phi_j(r_j) = r_j^2$ for $j = 1, 3, 5, \dots$, and r_j otherwise.
Voltage collapse	$[\delta_m, \omega, \delta, V]^\top$	Coordinatewise min-max scaling with frozen bounds from the original 10,000-sample training records; the scaled pair is multiplied through W_{stat} , whose standard deviation is set by the statistic-scaling hyperparameter	$\mathbf{z} = \mathbf{r}$.
Kuramoto network	Scalar order parameter r	Coordinatewise z-score over the five training conditions	$\mathbf{z} = [\mathbf{r}^\top, \tilde{r}, 1]^\top$, where $\tilde{r}$ is the standardized current order parameter.

S2.2 Conditioning path and test queries

Each training trajectory provides a measured pair $\mathbf{s}^{(m)} = (s_1^{(m)}, s_2^{(m)})$. A least-squares line

$$s_2 = as_1 + b \quad (3)$$

describes the training trend. The reported test and rate-sweep queries for all three systems use their corresponding measured-training fits. The physical parameter is never an input to the reservoir.

Training coordinates in Tables 3–5 are direct measurements from retained target trajectories. The reported test coordinates are the actual pairs supplied to the final SCDT. For Food and Voltage, the first test coordinate is obtained from interpolation, direct precritical measurement, or parameter-to-statistic extrapolation as specified below. The second coordinate follows the system-specific fit described below. The Kuramoto test pairs are treated in the same way after selecting asynchronous inter/pre trajectories.

S3 Conditioning statistics

S3.1 Standard deviation

For a scalar record $x_1, \dots, x_T$, the population standard deviation used in the code is

$$\sigma_x = \sqrt{\frac{1}{T} \sum_{n=1}^T (x_n - \bar{x})^2}, \quad \bar{x} = \frac{1}{T} \sum_{n=1}^T x_n. \quad (4)$$

The food-chain coordinate is σ_P and the voltage coordinate is σ_V .

S3.2 Lag-1 autocorrelation

The voltage experiment uses

$$\rho_1(V) = \frac{c_1}{c_0}, \quad c_0 = \frac{1}{T} \sum_{n=1}^T (V_n - \bar{V})^2, \quad c_1 = \frac{1}{T-1} \sum_{n=1}^{T-1} (V_n - \bar{V})(V_{n+1} - \bar{V}). \quad (5)$$

S3.3 Permutation entropy

The food-chain experiment uses normalized permutation entropy of $P(t)$ with embedding order $d = 3$ and delay $\tau = 1$. If $p(\pi)$ is the empirical probability of ordinal pattern π among the $d!$ possible patterns, then

$$H_{\text{perm}} = -\frac{1}{\log(d!)} \sum_{\pi} p(\pi) \log p(\pi), \quad d = 3, \quad \tau = 1. \quad (6)$$

The implementation uses the normalized estimator in the antropy package [5].

S3.4 Kuramoto order parameter and topology-aware correlations

Macroscopic synchronization is measured by

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|. \quad (7)$$

The SCDT dynamically predicts only the scalar $r(t)$. Node phases are used to construct the two static conditioning coordinates. Set $x_i(t) = \cos \theta_i(t)$ and let ρ_{ij} be the Pearson correlation between x_i and x_j over the retained measurement segment. The code uses $|\rho_{ij}|$. Nodes at or above the 90th percentile of degree form the hub set $\mathcal{H}$, and the remaining nodes form $\mathcal{L}$. The statistics are

$$S_{\text{int}} = \frac{1}{|\mathcal{H}|(|\mathcal{H}| - 1)} \sum_{\substack{i, j \in \mathcal{H} \\ i \neq j}} |\rho_{ij}|, \quad (8)$$

$$S_{\text{out}} = \frac{1}{|\mathcal{H}||\mathcal{L}|} \sum_{i \in \mathcal{H}} \sum_{j \in \mathcal{L}} |\rho_{ij}|. \quad (9)$$

These are experiment-specific, topology-aware regime coordinates motivated by correlation-based functional connectivity and hub synchronization studies [8, 9, 10]; they are not assumed to be universal early-warning indicators.

S4 Target dynamical systems

S4.1 Tri-trophic food chain

The state is $[R, C, P]^T$: resource, consumer, and predator. The exact equations implemented in the final experiment are

$$\dot{R} = R \left(1 - \frac{R}{K} \right) - \frac{x_c y_c C R}{R + R_0}, \quad (10)$$

$$\dot{C} = x_c C \left(\frac{y_c R}{R + R_0} - 1 \right) - \frac{x_p y_p P C}{C + C_0}, \quad (11)$$

$$\dot{P} = x_p P \left(\frac{y_p C}{C + C_0} - 1 \right). \quad (12)$$

The parameters are

$$x_c = 0.4, \quad y_c = 2.009, \quad x_p = 0.08, \quad y_p = 2.876, \quad R_0 = 0.16129, \quad C_0 = 0.5. \quad (13)$$

The carrying capacity K is varied only to generate and validate data. The model is a standard chaotic tri-trophic food-chain benchmark [3, 2].

Target trajectories are integrated by RK45. The solver output is evaluated at 0.1-time-unit intervals and every tenth value is retained, giving $\Delta t = 1$. For each candidate trajectory, 2,000 retained samples are discarded as target transient and 6,000 subsequent samples are retained. The initial condition is sampled independently as

$$R(0) \sim U(0.6, 1.0), \quad C(0) \sim U(0.15, 0.55), \quad P(0) \sim U(0.6, 0.7). \quad (14)$$

Up to 30 draws are attempted. A retained training record is accepted only when $\min P > 0.01$ and its final value satisfies $P > 0.01$.

S4.2 Voltage-collapse model

The four-dimensional state is $[\delta_m, \omega, \delta, V]^T$. The implemented Dobson–Chiang model is [4, 2]

$$\dot{\delta}_m = \omega, \quad (15)$$

$$M\dot{\omega} = -d_m\omega + P_m - E_m V Y_m \sin(\delta_m - \delta), \quad (16)$$

$$K_{qw}\dot{\delta} = -K_{qv2}V^2 - K_{qv}V + Q - Q_0 - Q_1, \quad (17)$$

$$TK_{qw}K_{pv}\dot{V} = K_{pw}K_{qv2}V^2 + (K_{pw}K_{qv} - K_{qw}K_{pv})V + K_{qw}(P - P_0 - P_1) - K_{pw}(Q - Q_0 - Q_1), \quad (18)$$

where

$$P = -E'_0 V Y'_0 \sin \delta + E_m V Y_m \sin(\delta_m - \delta), \quad (19)$$

$$Q = E'_0 V Y'_0 \cos \delta + E_m V Y_m \cos(\delta_m - \delta) - (Y'_0 + Y_m)V^2. \quad (20)$$

With $\theta_0 = 0$, the transformed source parameters are evaluated from

$$Y'_0 = Y_0 \sqrt{1 + (C/Y_0)^2 - 2(C/Y_0) \cos \theta_0}, \quad (21)$$

$$E'_0 = \frac{E_0}{\sqrt{1 + (C/Y_0)^2 - 2(C/Y_0) \cos \theta_0}}. \quad (22)$$

Table 2. Voltage-model parameters used in every simulation.

K_{pw}	K_{pv}	K_{qw}	K_{qv}	K_{qv2}	T	P_m	d_m
0.4	0.3	-0.03	-2.8	2.1	8.5	1.0	0.05
M	E_m	Y_0	Y_m	P_0	Q_0	P_1	(C, E_0)
0.01464	1.05	3.33	5.0	0.6	1.3	0.0	(3.5, 1.0)

The ODE is integrated by RK45 at $\Delta t = 0.05$, with maximum internal step 0.1, relative tolerance 10^{-4} , and absolute tolerance 10^{-6} . Stable training trajectories use

$$\delta_m(0) \sim U(0.17, 0.25), \quad \omega(0) \sim U(0, 0.05), \quad \delta(0) \sim U(0.05, 0.10), \quad V(0) \sim U(0.83, 0.87). \quad (23)$$

The first 500 time units (10,000 samples) are discarded. The next 3,000 time units (60,000 samples) form the training record; an additional 200-time-unit (4,000-sample) buffer is generated but is not included in the readout fit. The conditioning statistics are measured over all 60,000 training samples. For the displayed post-collapse ground truth, a delayed-collapse initial-condition region is used: $\delta_m(0) \sim U(0.17, 0.42)$, $\omega(0) = 0.025$, $\delta(0) = 0.10$, and $V(0) \sim U(0.70, 0.87)$.

S4.3 Degree-frequency correlated Kuramoto network

The target network obeys

$$\dot{\theta}_i = \omega_i + K \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i), \quad \omega_i = \frac{k_i}{\langle k \rangle}, \quad i = 1, \dots, N. \quad (24)$$

No additional $1/N$ normalization is applied to the coupling term. A fixed Barabasi–Albert graph [6] is generated with $N = 100$, attachment parameter $m = 3$, and network seed 0. It contains 291 edges, has mean degree 5.82, and has degree range 1–32. The degree-frequency correlation produces explosive synchronization in this variant of the Kuramoto model [11, 7].

The ODE is integrated by RK45 with $\Delta t = 0.05$ and relative and absolute tolerances 10^{-7} . For every training condition, the target is evolved for 185 time units. The first 70 time units (1,400 sampling intervals) are discarded as target transient, leaving 115 time units (2,301 samples, including both endpoints) for statistics and RC training. All five selected training trajectories use target seed 26 and remain asynchronous. The final inter, pre-synchronization trajectories displayed in Fig. 9 also use target seed 26. The post query in Fig. 8 is assessed by a separate 20-window SCDT check, not by a displayed seed-26 post trajectory. Selected-point GT checks use independent phase seed 26, rather than the forward-continuation initialization of the GT rate curve. The 10 highest-degree nodes form $\mathcal{H}$ and the other 90 nodes form $\mathcal{L}$.

The target forward continuation spans $K \in [0.15, 0.35]$ in steps of 0.01; the final rate figure displays its 12 GT points in $K \in [0.19, 0.30]$. The SCDT is evaluated at 18 uniformly spaced conditioning settings over the same calibration-label interval. Each of 20 target realizations starts from independent random phases at $K = 0.15$. At each coupling, the target is integrated for 500 time units, and synchronization is evaluated over the final 150 time units. The final phase state is then reused as the initial condition at the next value on the increasing- K grid. No synchronized-branch initialization is imposed. The coupling value and continuation state are not supplied as SCDT inputs.

S5 Training and test coordinates

Table 3. Food-chain coordinates. The SCDT receives the two statistics, not K . Training statistics are direct measurements from the selected ODE trajectories.

Role	K	σ_P	H_{perm}	Target seed	Coordinate source
Train	0.9740	0.121337	0.505563	8	Direct target measurement
Train	0.9780	0.123094	0.506485	5	Direct target measurement
Train	0.9800	0.125328	0.508786	1	Direct target measurement
Train	0.9820	0.126225	0.510812	0	Direct target measurement
Train	0.9880	0.128283	0.512399	8	Direct target measurement
Inter	0.9850	0.127254	0.511310	–	Conditioning-path query
Pre	0.9970	0.133339	0.517649	–	Conditioning-path query
Post	1.0001	0.134923	0.519299	–	Conditioning-path query

Table 4. Voltage-collapse coordinates. The SCDT receives the two statistics, not Q_1 .

Role	Q_1	σ_V	$\rho_1(V)$	Coordinate source
Train	2.9896800	0.0316339	0.9834461	Direct target measurement
Train	2.9897050	0.0317339	0.9835333	Direct target measurement
Train	2.9897300	0.0319752	0.9838649	Direct target measurement
Train	2.9897550	0.0322487	0.9841907	Direct target measurement
Train	2.9897800	0.0325905	0.9846133	Direct target measurement
Inter	2.9897475	0.0321755	0.9841014	Direct σ_V ; projected ρ_1
Pre	2.9897840	0.0326679	0.9847096	Direct σ_V ; projected ρ_1
Post	2.9898270	0.0330095	0.9851316	Projected near-critical extrapolation query

For Voltage, the five displayed training pairs are direct measurements from the 60,000-sample records and exactly match the normalized statistics stored with the trained reservoir. Statistic normalization retains the coordinatewise minimum and maximum from the original five 10,000-sample training records. These bounds are reused for the final 60,000-sample training records and all test and rate queries, without refitting the normalization. Their fitted conditioning relation is $\rho_1 = 1.2351899093 \sigma_V + 0.9443585599$. The inter, pre, and post test queries, as well as every SCDT rate-sweep query, use this same relation. The rate-sweep inputs are 35 uniformly spaced σ_V values, and the displayed rate axis is aligned by the Fig. 7(b) raw regression described below. The measured-training fit $\sigma_V = 9.7123093821 Q_1 - 29.0051462885$ is used only to assign original query labels for selecting the training warm-up record and conditioning schedule; Q_1 itself is used only for target generation and axis alignment. Refitting the readout on the 299,495 effective training pairs reproduced the stored readout weights exactly; the training inputs and reservoir weights were not changed.

Table 5. Kuramoto coordinates. Column order matches the conditioning vector used by the code, $[S_{\text{out}}, S_{\text{int}}]$.

Role	K	S_{out}	S_{int}	Target seed	Coordinate source
Train	0.2180	0.056196	0.183096	26	Direct target measurement
Train	0.2210	0.057167	0.203595	26	Direct target measurement
Train	0.2260	0.056519	0.208676	26	Direct target measurement
Train	0.2290	0.058910	0.216242	26	Direct target measurement
Train	0.2320	0.060497	0.227692	26	Direct target measurement
Inter	0.2300	0.059169325	0.218284892	26	Training-regression query
Pre	0.236042416	0.060820326	0.231408065	26	Training-regression query
Post	0.2400	0.061901678	0.240003312	–	Training-regression query

The Kuramoto training-coordinate fits are $S_{\text{out}} = 0.273235303842 K - 0.003674794901$ and $S_{\text{int}} = 7.948614050530 S_{\text{out}} - 0.252029235654$. Selected test pairs follow these fits and are the actual coordinates supplied to the SCDT. During the pre search, S_{int} is scanned and S_{out} is obtained by inverting the second fit; the first fit then identifies the corresponding target-query K . These supplied test coordinates are not newly measured raw statistics of the test trajectories. The selected pre satisfies $0.227692011 < S_{\text{int}} = 0.231408065 < 0.231968081$, placing it between the maximum training S_{int} and the displayed coarse-grid SCDT crossing. Its original target-query $K = 0.236042416$ exceeds the largest training $K = 0.232$. Its $S_{\text{out}} = 0.060820326$ also exceeds the maximum raw training $S_{\text{out}} = 0.060497060$. All 20 training-window SCDT reproductions remain asynchronous at this pre; the independently integrated seed-26 GT trajectory is also asynchronous. This is a single-realization GT check, not a 20-seed GT ensemble result.

For Food and Voltage inter and pre queries, the requested conditioning coordinate is held fixed during warm-up and autonomous prediction. For Kuramoto, warm-up pairs the selected training trajectory with its own training conditioning coordinate; the requested query coordinate is applied from the first autonomous step and held fixed thereafter. Food uses the last-training conditioning for the first 500 autonomous steps and then ramps to the post coordinate over 50 steps. Voltage and Kuramoto use a constant requested conditioning coordinate throughout the autonomous rollout. Dynamic warm-up samples are taken only from a selected training trajectory for all three systems. Warm-up samples initialize the reservoir state but are excluded from every autonomous evaluation window reported below. **The Food rate sweep uses a single input schedule at every conditioning setting: the requested coordinate is held constant during the 500-sample training-trajectory warm-up and all 1,000 autonomous samples. There is no critical-parameter-dependent switch, autonomous hold, or ramp in the rate sweep. The separately displayed post trajectory retains the hold/ramp protocol described above.**

S6 Sequence lengths and rollout protocol

Table 6. Complete sequence accounting. A “step” denotes one retained sample interval after ODE resampling.

Item	Food chain	Voltage collapse	Kuramoto network
Sampling interval	$\Delta t = 1$	$\Delta t = 0.05$	$\Delta t = 0.05$
Target transient discarded	2,000 steps = 2,000 time units	500 time units = 10,000 steps	70 time units = 1,400 intervals
Retained target record per training condition	6,000 steps	3,000 time units = 60,000 training samples, plus a 4,000-sample buffer	115 time units = 2,301 samples
Window used to calculate conditioning statistics	First 2,000 samples of retained $P(t)$	All 60,000 samples of the training-window $V(t)$	All 2,301 retained phase samples
Training input/target pairs before washout	1,999 per condition	59,999 per condition	2,300 per condition
Training washout	Updates $n \leq 100$ excluded	First 100 updates excluded = 5 time units	First 80 updates excluded = 4 time units
Readout pairs retained	1,898 per condition; 9,490 total	59,899 per condition; 299,495 total	2,220 per condition; 11,100 total
Prediction target seed buffer	Selected inter/pre: final 500 samples from a training trajectory; rate: sampled 500-sample training windows	100 samples from the nearest selected training trajectory; endpoint 9,500 for inter/pre and 26,375 for the displayed post case; rate: 50 training-window endpoints per Q_1	Displayed inter/pre: member 0 among 20 evenly spaced 60-sample windows from the nearest training trajectory; rate curve: all 20 windows
Samples actually fed during prediction warm-up	500 steps	100 steps = 5 time units	60 steps = 3 time units
Autonomous prediction horizon	Selected trajectories: 2,000 steps; rate: 1,000 steps ($\Delta t = 1$)	Inter/pre and statistical reconstruction: 10,000 steps = 500 time units; post and rate: 2,000 steps = 100 time units	10,000 steps = 500 time units for both selected-point checks and the rate curve; first 2,000 steps shown in Fig. 9
Post-query conditioning schedule	Last-training conditioning held for 500 autonomous steps, followed by a 50-step linear ramp to the post coordinate	Requested post coordinate fixed throughout the autonomous rollout	Post coordinate fixed throughout the autonomous rollout
Displayed trajectory length	2,000 steps	1,000 steps for inter/pre; first 200 of 2,000 generated steps for the post comparison	2,000 steps

S7 Reservoir hyperparameters and constraints

Table 7. Final reservoir settings. “Density” follows the exact mask semantics of each implementation; this avoids interpreting the configuration field `sparsity` in the opposite direction.

Parameter	Food chain	Voltage collapse	Kuramoto network
Reservoir nodes N_r	1,000	900	2,000
Recurrent-matrix construction	Sparse random matrix with requested pre-symmetrization density $113/1000 = 0.113$; matrix is symmetrized	Gaussian dense draw masked by $U > 0.91$; approximately 9% retained	Gaussian draw with entries zeroed for $U > 0.972323118$; approximately 97.23% retained
Spectral radius	1.5	0.5346534903	1.153870539
Leakage α	0.01	0.1426937444	0.09044233509
Ridge coefficient β	0.00183011	0.08711372385	10.0
Dynamic-input scaling	5.0	2.566087561	0.04672287231
Statistic-input scaling	(0.1, 0.1) after z-scoring	0.125 in W_{stat}	0.4542375643 in W_{stat}
Statistic offset	(−1.631799943, 0.0517568555)	None after min-max scaling	None after z-scoring
Reservoir seed	11	42	23
Output constraint	Negative state components clipped to zero; rollout terminated at $P < 0.01$ by filling the remaining output with zero	Componentwise clipping to [−0.10, −1.50, −0.30, 0.01] and [0.70, 1.20, 0.40, 1.10]	Standardized output clipped to ± 5.716762690 , then inverse-transformed and clipped to $r \in [0, 1]$
Prediction mode	Full-state closed loop	Full-state closed loop	Scalar r -only closed loop

The final Kuramoto checkpoint was obtained by exploratory searches over five-point training sets and $N_r = 2000$ reservoir settings, followed by full-horizon, 20-window validation and ridge-coefficient refinement. Selection used the frozen GT transition to favor an earlier SCDT crossing together with asynchronous training/pre queries and a synchronized high-coordinate response. These are exploratory model-selection results, not independent held-out validation of the critical-point estimate. A separate $K = 0.240$ post query has subsequently been checked with all 20 training warm-up windows. The standardized-output clipping ceiling maps to $r = 0.550513278$ with the stored signal normalizer; synchronization outcomes use this constrained output and do not imply reproduction of the GT post-branch amplitude near $r = 1$. The inter and pre trajectories in Fig. 9 use the saved final checkpoint; the rate curve in Fig. 10 uses the same checkpoint and training coordinates. For the displayed inter and pre trajectories, the nearest training records are $K = 0.229$ and $K = 0.232$, respectively. The first of the 20 training windows is asynchronous in each case and is the window plotted in Fig. 9.

S8 Outcome criteria and rate curves

S8.1 Selected-trajectory classification

The qualitative selected-point checks use the following definitions:

- **Food-chain survival:** a 2,000-step trajectory survives when the mean predator population over the final 100 steps is greater than 0.05. The first sample with $P < 0.05$ is recorded as the event step.
- **Voltage survival:** a 2,000-step trajectory survives when every autonomous voltage sample is finite and $V > 0.30$ throughout the rollout. The first sample with $V \leq 0.30$ is recorded as the event time.
- **Kuramoto synchronization:** the evaluation tail is the final 30% of a 10,000-step prediction, namely 3,000 samples or 150 time units. A trajectory is synchronized when at least 80% of those samples satisfy $r \geq 0.5$. Equivalently, at least 2,400 of the final 3,000 samples, corresponding to at least 120 time units in total, must pass the threshold. The passing samples need not be contiguous.

For the Kuramoto inter/pre checks, the autonomous prediction lasts 500 time units after a 3-time-unit training-trajectory warm-up. Figure 9 displays only its first 100 time units. Classification uses the final 3,000 autonomous samples at $t = 350.05, 350.10, \dots, 500.00$, not the displayed first 100 or the discarded 70-time-unit target transient. For the independent selected-point GT, the matching evaluation samples are at absolute target times 420.05, 420.10, $\dots$, 570.00 after the 70-time-unit transient.

For the final Kuramoto rate curve, the same threshold and occupancy fractions are applied to a 10,000-sample autonomous prediction (500 time units). Its evaluation tail is the final 3,000 samples (150 time units), and synchronization requires at least 2,400 of those samples (120 time units in total) to satisfy $r \geq 0.5$. Warm-up samples are excluded from both evaluation windows.

S8.2 Rate-sweep grids and ensembles

For the Kuramoto rate MAE, the 18-point SCDT curve is linearly interpolated at the 12 directly evaluated GT coordinates on the common statistic axis. The GT rates are not interpolated or recomputed for this comparison. These are distinct conditioning grids, not different ensemble sizes. For Food and Voltage, GT and SCDT are evaluated at matching abscissae on the common statistical axis. The 22 Food supplied values are uniformly spaced over $[0.119237869, 0.149162935]$ with spacing 0.001425003; the 35 Voltage values cover $[0.030830874, 0.034524236]$ with spacing 0.000108628. The same panel-(b) regression maps the corresponding GT parameter grid onto these coordinates. MAEs compare the paired measured rates directly, without interpolation. All 440 Food and 1,750 Voltage SCDT member rollouts were regenerated with the frozen reservoirs and training-only warm-up protocols; every evaluated point is retained. **All 440 Food rollouts use the same constant-requested-coordinate rule. The frozen reservoir, requested coordinates, training source records, 20 warm-up windows per setting, and survival criterion are unchanged.**

The final Food GT rate retains the 22 directly integrated conditions in $K \in [0.975, 1.010]$ from the previous 31-condition GT calculation; it is not an interpolation of stored survival fractions. Initial conditions are the fixed 20-member subset selected with seed 20260922 from the original baseline at the nearest original K . No initial conditions are filtered by their outcome in the new integrations. After a 500-time-unit target transient, 1,000 samples at $\Delta t = 1$ are evaluated. The survival mean uses the final 500 samples. At each matched statistical coordinate, the 20 SCDT runs use the same evaluation length, sampling interval, tail, and threshold, excluding their 500-sample training-trajectory warm-up. All rate-query entropy values use the same training $\sigma_P - H_{\text{perm}}$ fit as the selected test queries. This does not equate GT random initial conditions with SCDT warm-up states.

The GT evaluation samples are $t = 500, 501, \dots, 1499$, with the mean calculated at $t = 1000, \dots, 1499$. For SCDT, autonomous sample indices $0, \dots, 999$ are evaluated and indices $500, \dots, 999$ enter the mean; no teacher-forced samples

Table 8. Final rate-curve protocol and provenance.

Item	Food chain	Voltage collapse	Kuramoto network
GT/calibration-parameter grid	GT and SCDT calibration: $K \in [0.975, 1.010]$, 22 equally spaced points each	$Q_1 \in [2.989620, 2.989960]$, 35 points at 10^{-5} spacing	GT: 12 points, $\Delta K = 0.01$; SCDT calibration labels: 18 points, $\Delta K = 0.11/17 \approx 0.006470588$, both in $[0.19, 0.30]$
Representative lower-axis statistic	σ_P	σ_V	S_{int}
GT ensemble	20 fresh ODE integrations per final-grid condition, using fixed baseline initial-condition subsets	50 fresh ODE integrations per final-grid condition, using the original random-initial-condition family	20 forward-continuation ODE realizations per point
SCDT ensemble	20 closed-loop rollouts per point	50 closed-loop rollouts per point	20 closed-loop rollouts per point, initialized by distinct training-trajectory windows
GT/SCDT evaluation horizon	1,000 samples, $\Delta t = 1$; 1,000 time units after initialization	2,000 samples, $\Delta t = 0.05$; 100 time units	10,000 samples, $\Delta t = 0.05$; 500 time units
Rate rule	Final-500-step mean $P > 0.05$	All 2,000 predicted voltage samples finite and $\min V > 0.30$	Final 3,000 of 10,000 autonomous samples; at least 80% with $r \geq 0.5$
Reported GT–SCDT MAE	21.1364 percentage points	17.7143 percentage points	0.03560606 in rate fraction, on the 12 GT coordinates

enter either SCDT evaluation window. The GT rates use the first 1,000 samples of the saved ODE records. All 22 SCDT conditions were freshly evaluated over 1,000 autonomous samples; the saved member trajectories reproduce every reported survival decision. Selected inter/pre/post trajectory checks retain their separate 2,000-sample horizon and final-100-sample criterion; the displayed post rollout also retains its distinct conditioning hold/ramp protocol.

For Voltage, the 50 directly integrated target trajectories at each of the 35 displayed Q_1 values are retained. Initial conditions retain the original baseline’s distribution and seed convention, using the nearest original baseline row to identify the seed family, with no outcome-based resampling. Both GT and SCDT use a 100-time-unit evaluation horizon at $\Delta t = 0.05$. GT samples are evaluated from $t = 0$ through 99.95, so early threshold crossings count; no initial 25-time-unit GT interval is discarded. SCDT evaluation starts after its 100-sample training-trajectory warm-up. A sampled $V \leq 0.30$ establishes non-survival even if integration later truncates. All 906 truncated GT runs contain such a sampled violation, leaving no unresolved decisions. The frozen consistent-final SCDT is evaluated afresh at all 35 uniformly spaced input statistics, with 50 member rollouts per point; their saved trajectories verify every reported member decision.

Kuramoto target and SCDT rates use different but explicit initialization protocols. The target uses 20 independent forward continuations. At each supplied coordinate, the SCDT is reset and warmed up with 20 evenly spaced 60-sample windows from the nearest training trajectory; the warm-up samples are excluded from the 10,000-step autonomous evaluation. These windows are different reservoir initializations, not independent target-system seeds or independent reservoir-weight realizations.

The red dashed lines in Figs. 4(a), 7(a), and 10(a) mark the respective SCDT 50% crossings. For Food, the crossing is at the supplied $\sigma_P = 0.134200402$, whose Fig. 4(b) axis-aligned label is $K = 0.992500000$. The target 50% crossing is at actual $K = 1.000000000$, mapped to $\sigma_P = 0.140612916$ by the same Fig. 4(b) regression. This actual-parameter crossing equals the physical reference $K_c = 1.0000$ on the displayed GT grid. The separate selected Food post query has supplied $\sigma_P = 0.134923$, beyond the SCDT rate crossing; its target is generated at $K = 1.0001$ and collapses under the selected-trajectory protocol. Its supplied coordinate is not a direct measurement of the Fig. 4(b) regression at that target K .

The final Voltage sweep uses the measured-training conditioning relation given above at all 35 supplied statistic values. At each value, the SCDT is independently reset and evaluated from 50 training-trajectory warm-up windows, using the 2,000-step survival rule. The empirical SCDT 50% crossing is $\sigma_V = 0.032734585$, aligned with $Q_1 = 2.989795250$ using Fig. 7(b). The target 50% crossing is at actual $Q_1 = 2.989825135$, mapped to $\sigma_V = 0.033059222$ by Fig. 7(b). The matched-statistic GT–SCDT rate MAE is 17.7143 percentage points. Figures 5–6, the selected test queries, and this rate curve use the same measured-training σ_V – ρ_1 relation for the second input coordinate. The separate raw target measurements in Fig. 7(b) do not serve as SCDT inputs.

The training-parameter mapping identifies the training warm-up record; it does not determine the displayed upper axis. For the Food rate curve, original query labels do not select a conditioning schedule: the requested coordinate is constant at every setting. The final Food supplied interval corresponds to original query labels $[0.969414215, 1.027956992]$; the Voltage interval corresponds to $[2.989605872, 2.989986149]$. These original labels differ from the displayed calibration

intervals. The requested first statistic and its training-fit second statistic are the actual SCDT inputs.

For Kuramoto, the validation-axis calibration is the raw-panel regression $f_b(K) = 1.057809579698 K - 0.020350615386$ from Fig. 10(b). GT points are placed at $f_b(K_{GT})$; SCDT points retain their actual supplied S_{int} ; the upper axis displays f_b^{-1} . The 18 SCDT input settings are uniformly spaced over $S_{\text{int}} \in [0.180633205, 0.296992259]$, corresponding to the calibration-label interval $[0.19, 0.30]$ with step $0.11/17$. The actual statistic spacing is $\Delta S_{\text{int}} = 0.006844650$. The original training-based query K used to select a warm-up record is different from this calibration label. It is obtained by inverting the two training fits, and spans $[0.212663708, 0.266239914]$ for these 18 settings. The selected pre has calibration label 0.238000000 , while its independent target trajectory is generated at $K = 0.236042416$.

S8.3 Direct target-statistic checks in panels (b)

The points in Figs. 4(b), 7(b), and 10(b) are *direct target measurements*, not the regression-produced conditioning values on the lower axis of panels (a). Their dashed lines are least-squares regressions of the additional target measurements displayed in the respective panels. These raw-statistic regressions describe the local parameter–statistic trends and are also the validation-axis mappings in all three corresponding panels (a): GT abscissae are $f_b(p_{GT})$, SCDT abscissae remain the actual supplied statistic, and the upper axis is f_b^{-1} . The raw checks are separate from the five training trajectories and do not enter the reservoir.

For Fig. 4(b), 30 surviving target trajectories from seed 43 are shown in the $K \in [0.974, 0.990]$ neighborhood. Their predator standard deviations use 2,000 target samples each. For Fig. 7(b), 40 surviving seed-43 target trajectories are shown in the $Q_1 \in [2.989675, 2.989784]$ neighborhood; their voltage standard deviations use 10,000 samples each.

For Fig. 10(b), each of 40 conditions in $K \in [0.190, 0.225]$ reports the mean of 20 direct target measurements of S_{int} . Each measurement uses the final 150 time units of a 500-time-unit forward continuation. Only conditions for which all 20 target realizations remained asynchronous were eligible; the 40 displayed conditions span the precritical interval. Raw statistics can differ from supplied regression coordinates because they are measured on different realizations and finite windows. These restricted panels show local trends, not global linearity across the full rate sweep.

S8.4 Critical values and statistical-axis alignment

The upper physical-parameter axis in the rate figures is a validation aid. System-specific mappings use exactly the raw-measurement least-squares regressions displayed in panels (b). For Food, $f_b(K) = 0.855001902518 K - 0.714388986243$; for Voltage, $f_b(Q_1) = 10.862830916202 Q_1 - 32.444905689842$. The Kuramoto mapping is given above. These local fits are extrapolated where the rate grid extends beyond the raw measurement intervals. They do not replace the training stat-space fits that set the second SCDT input coordinate, or alter the supplied first coordinate. The upper-axis parameter labels are calibration labels and are never supplied to the SCDT. Table 9 reports the empirical target and SCDT 50% crossings; the collapse-model physical references are listed separately. We denote the mapped target crossing by $s_c^{(GT)}$ and the empirical SCDT crossing by $\hat{s}_c^{(SCDT)}$; the two quantities need not be numerically identical.

Table 9. Critical values calculated from the rate files used in the main figures.

System	Physical reference and empirical target crossing	Empirical SCDT 50% rate crossing	Interpretation
Food chain	Physical reference: $K_c = 1.0000 \mapsto \sigma_P = 0.140613$; target 50% crossing: $K = 1.000000000 \mapsto \sigma_P = 0.140612916$	Calibration $K = 0.992500000$, $\sigma_P = 0.134200402$	Linear interpolation between adjacent SCDT survival-rate samples.
Voltage collapse	Physical reference: $Q_{1,c} = 2.9898256 \mapsto \sigma_V = 0.033064272$; target 50% crossing: $Q_1 = 2.989825135 \mapsto \sigma_V = 0.033059222$	Calibration $Q_1 = 2.989795250$, $\sigma_V = 0.032734585$	Linear interpolation between adjacent SCDT survival-rate samples.
Kuramoto network	Target 50% crossing: $K = 0.240 \mapsto S_{\text{int}} = 0.233523684$	Calibration $K = 0.238529412$, $S_{\text{int}} = 0.231968081$	Crossings from the final 500-time-unit, 20-member rate curves.

These values should not be conflated. The mapped target crossing places the empirical target outcome on the common statistical axis. Physical references for the two collapse models have a separate definition. The SCDT 50% crossing is an empirical estimate obtained from its finite ensemble and discrete rate grid.

For each rate curve, the reported crossing is obtained from the two adjacent sampled points that bracket a rate of 0.5. If their coordinates and rates are (s_1, p_1) and (s_2, p_2) , the crossing is

$$s_{50} = s_1 + \frac{0.5 - p_1}{p_2 - p_1}(s_2 - s_1). \quad (25)$$

Thus the quoted SCDT critical coordinate is the linearly interpolated 50% crossing, whereas the target-aligned empirical value is the target 50% crossing mapped onto the same statistical axis. For the same Kuramoto checkpoint, a separate 56-condition uniform sweep with calibration spacing 0.002 gives $S_{\text{int},50} = 0.232465874$ (calibration $K = 0.239$). Figure 10 and Table 9 use only the displayed 18-condition SCDT grid and the unchanged 12-condition GT grid. The SCDT crossing is a coarse interpolation, not an additional simulation at the crossing coordinate. The new pre check is reported separately and is not inserted into that 18-condition curve. With the model, range, and selected pre fixed, candidate uniform SCDT grids of 13–17 points were checked at their transition brackets; none placed the displayed SCDT crossing both above the pre and below the frozen GT crossing. The 18-point grid was the first to satisfy this display-order criterion and was evaluated over its full range. This grid-selection result does not make the interpolated crossing a resolution-independent critical value.

S9 Selected-point results and statistical self-consistency

S9.1 Qualitative outcomes

Table 10. Outcomes of the displayed selected trajectories. Where reported, MAE is computed pointwise over the 2,000-step autonomous segment.

System	Role	GT outcome	SCDT outcome	Trajectory result
Food chain	Inter	Survived	Survived	MAE = 0.126414
	Pre	Survived	Survived	MAE = 0.156667
	Post	Collapsed	Collapsed	GT event step 411; SCDT event step 629
Voltage collapse	Inter	Survived	Survived	MAE = 0.033482
	Pre	Survived	Survived	MAE = 0.032724
	Post	Threshold crossed	Threshold crossed	GT event at 7.90; SCDT event at 8.85 time units
Kuramoto network	Inter	Asynchronous	Asynchronous	MAE = 0.052154
	Pre	Asynchronous	Asynchronous	MAE = 0.109709

The outcome classification, rather than pointwise agreement over a long chaotic rollout, is the primary test for the transition experiments. All eight displayed GT/SCDT trajectory pairs have the expected qualitative state. The Kuramoto post coordinate is shown in the stat-space diagram and included in an additional 20-window SCDT check, but no single post trajectory is displayed in Fig. 9. The GT rate at $K = 0.240$ is 0.50. The SCDT grid instead has rates 0.00 at calibration label $K = 0.235294118$ and 1.00 at $K = 0.241764706$; $K = 0.240$ is not an independently evaluated SCDT grid point. The additional selected post check and these rate-curve settings should not be described as a matched selected GT/SCDT post pair.

The displayed Voltage post condition is $Q_1 = 2.989827$, only 1.4×10^{-6} above the physical reference $Q_{1,c} = 2.9898256$. The GT uses collapse-basin initial-condition seed 28 and the SCDT uses the nearest training trajectory ($Q_1 = 2.989780$) with a 100-sample warm-up ending at sample 26,375. The GT first satisfies $V \leq 0.30$ at autonomous step 158 ($t = 7.90$), and the SCDT first satisfies it at step 177 ($t = 8.85$). The SCDT remains below the threshold through the end of the displayed 200-step window. It can recover in a longer 2,000-step rollout, so the displayed event is a threshold crossing rather than proof of permanent collapse.

S9.2 Supplied versus reproduced statistics

For Food and Voltage, the statistics were remeasured from the closed-loop SCDT trajectories at the inter and pre conditions. This test asks whether a supplied conditioning coordinate is approximately reproduced by the generated dynamics; the statistics are not imposed as hard constraints in Eq. (1).

In the supplied-versus-reproduced standard-deviation panels (Figures 3(a) and 6(a) of the main manuscript), each candidate point denotes one conditioning setting and one deterministic closed-loop rollout with the fixed final reservoir seed. It is neither a different ODE seed nor an ensemble average. Food uses 40 noncollapsed conditioning settings and measures the reproduced standard deviation over all 2,000 autonomous samples ($\Delta t = 1$). The fixed plotting window displays 36 of these 40 Food settings. Voltage evaluates 40 noncollapsed conditioning settings, of which 39 fall inside the fixed plotting window,

and measures each over all 10,000 autonomous samples ($\Delta t = 0.05$, 500 time units). The 500-sample Food warm-up and the 100-sample Voltage warm-up are excluded from these evaluation windows, so no teacher-forced samples contribute to the reproduced statistics.

Table 11. Self-consistency of the selected noncollapsed inter/pre trajectories. Relative error is (reproduced – supplied)/supplied.

System	Role	Supplied σ	Reproduced σ	Error	Supplied stat 2	Reproduced stat 2	Error
Food	Inter	0.127254	0.129448	+1.724%	0.511310	0.515384	+0.797%
Food	Pre	0.133339	0.132420	−0.689%	0.517649	0.508985	−1.674%
Voltage	Inter	0.0321755	0.0322780	+0.319%	0.984101	0.984219	+0.0119%
Voltage	Pre	0.0326679	0.0328875	+0.672%	0.984710	0.985125	+0.0421%

For Kuramoto, the dynamic SCDT output contains only $r(t)$ and does not contain the 100 node phases. The pair ($S_{\text{out}}, S_{\text{int}}$) therefore cannot be remeasured from the SCDT output without extending the readout to the node level.

S10 Reproducibility notes

1. Five pre-transition target conditions are used for every system. The physical parameter values label the numerical data-generation runs but are absent from the reservoir input.
2. Statistics are calculated once per retained target segment and are held constant while that segment is processed by the reservoir.
3. The readout is trained one step ahead and is then evaluated in closed loop. A low one-step training error is not used as a substitute for the long-horizon survival or synchronization test.
4. Food and Voltage output clipping prevents numerical divergence from dominating the outcome test. The clipping bounds are listed in Table 7 and are part of the final protocol.
5. Collapse trajectories are excluded when measuring the target-only representative-statistic mapping used for the upper/lower axis alignment. This filtering affects only the validation alignment, not the five SCDT training trajectories.
6. The measured-training stat-space linearity coefficients are $R^2 = 0.96918$ for Food, $R^2 = 0.99912$ for Voltage, and $R^2 = 0.75565$ for Kuramoto.

S11 References

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